Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2024

Course: B.Tech | *Time: 03 Hours*
Code: 100502 | *Subject: Control Systems* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): [2 x 7 = 14]

(a) A unity feedback system has $G(s) = \frac{k}{s(s+3)}$. What should be the value of k to achieve a steady-state error of 0.1 for a unit ramp input?

(i) 3
(ii) 10
(iii) 30
(iv) 100

(b) A lead compensator has transfer function $G(s) = \frac{s+4}{(s+1)}$. What is the maximum phase lead it can

(i) 45°
(ii) 90°
(iii) 60°
(iv) 30°

(c) The state-space model $\dot{x} = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$ has eigenvalues:

(i) -1, -2
(ii) -2, -3
(iii) $-1 \pm j1$
(iv) -1, -1

(d) If the system is described by $\frac{C(S)}{R(S)} = \frac{S+3}{S^3+4S+5}$, determine if it is stable

(i) Stable
(ii) Marginally stable
(iii) Unstable
(iv) Cannot determine

(e) For a second-order system with $\zeta = 0.5$ and $\omega_n = 4$ rad/s. Compute the peak time.

(i) $\frac{\pi}{4}$
(ii) $\frac{\pi}{2}$
(iii) $\frac{\pi}{2\sqrt{3}}$
(iv) $\frac{\pi}{\sqrt{3}}$

(f) Find steady-state value using the final value theorem for $Y(S) = \frac{2}{S^2+4}$

(i) 0
(ii) ∞
(iii) 2
(iv) -1 to +1

(g) A PID controller is used when the system requires:

(i) High accuracy and fast response
(ii) Poor robustness
(iii) High overshoot
(iv) Slow correction

(h) For the characteristic equation $s^4 + 4s^3 + 6s^2 + 4s + k = 0$, determine the value of k for marginal stability.

(i) 4
(ii) 6
(iii) 0
(iv) 8

(i) The breakaway point of the root locus of $G(s)H(s) = \frac{k}{s(s+1)(s+4)}$ lies at:

(i) -2
(ii) -1.5
(iii) -3
(iv) -2.5

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(j) The Nyquist plot of $H(s) = \frac{1}{(s+1)}$ encircles the (-1 +j0) point how many times?

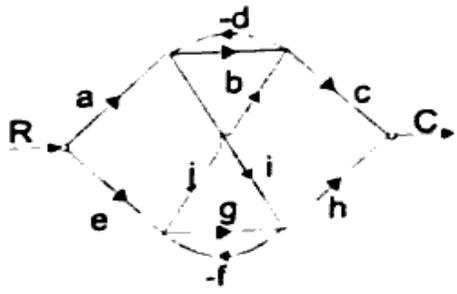
(i) 0
(ii) 1
(iii) -1
(iv) ∞

Q.2 (a) A unity feedback system has open open-loop transfer function

$$G(s) = \frac{k}{s(s+4)(s^2+4s+20)}$$

Sketch the root locus plot with 'k' as a variable parameter.

(b) Determine the transfer function from the given SFG



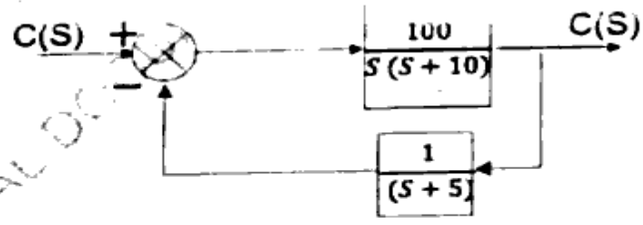
Q.3 Sketch a Bode plot for a unity-feedback system characterized by the loop transfer function and comment on stability.

$$G(s) = \frac{1500(1+0.2s)(1+0.025s)}{s^3(1+0.001s)(1+0.005s)}$$

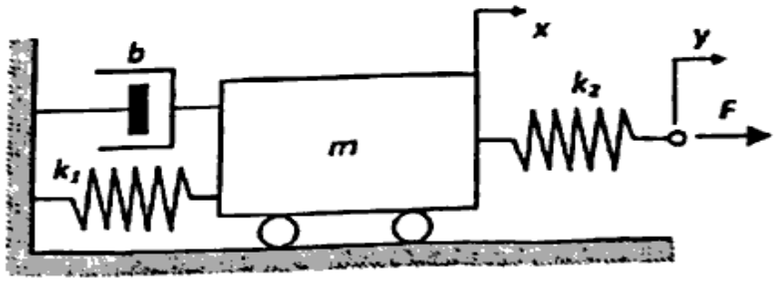
Q.4 (a) Find the number of poles in the left half-plane, right half-plane, and on the jω-axis for the system having the characteristic equation

$$p(s) = s^8 + 3s^7 + 10s^6 + 24s^5 + 48s^4 + 96s^3 + 128s^2 + 192s + 128$$

(b) A non-unity feedback system is shown below. Determine the steady-state error for a unit step input.



Q.5 In the mechanical system shown below, the force F is the input to the system and the displacement x, measured from the equilibrium position, is the output of the system, which is initially at rest.



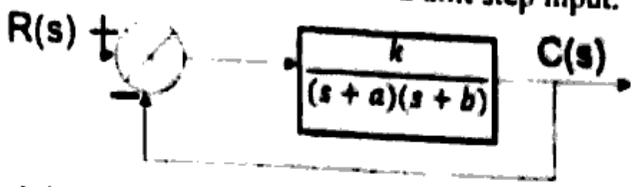
(a) Obtain the transfer function $\frac{X(S)}{F(S)}$

(b) Suppose m=1 kg, $k_1 = 2N/m$, $k_2 = 8N/m$, b = 5 Ns/m and F = 5 N, find the displacement x(t).

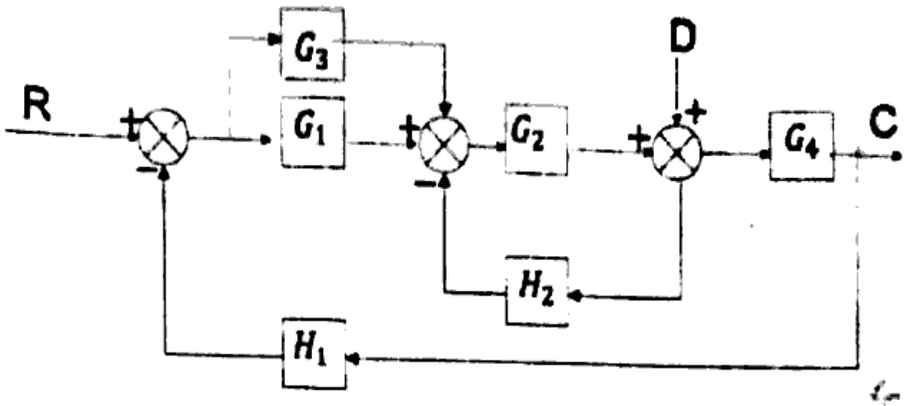
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(a) Find the sensitivity of the steady-state error to changes in parameter 'k' and parameter 'a' for the system shown below with a unit step input. [7]



(b) Find C/R, C/D, and the total output of the system whose block diagram is shown below [7]



Q.7 (a) A unity feedback system has an open-loop transfer function $G(s) = \frac{s+2}{(s+1)(s-1)}$ [7]
Use the Nyquist criteria to determine the system stability in the closed-loop configuration.

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(b) A unity feedback system has an open-loop transfer function: [7]

$$G(s) = \frac{k}{s(s+2)}$$

Design a lead compensator such that the phase margin is increased to 50° and the gain crossover frequency is approximately 4 rad/sec.
Find:
a) The required phase lead.
b) The lead compensator transfer function.
c) The new gain k of the system.

Q.8 (a) Check the observability of the system given below [7]
$\dot{x}_1 = x_2$; $\dot{x}_2 = -2x_1 - 3x_2 + u$; and $y = x_1 + x_2$

(b) Find the transfer function from the below state-space equation [7]

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} x + \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} u \quad \text{, and } y = [1 \quad 0 \quad 0]x$$

Q.9 Write short notes on ***any two*** of the following: [7 x 2 = 14]
(a) Define phase margin and gain margin. How do they relate to system stability?
(b) List and briefly explain the effects of proportional, integral, and derivative controllers on system performance.
(c) Define controllability and observability. Why are they important in state-space analysis?
(d) Define time response specifications such as rise time, peak time, and settling time for a second-order system.

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